# Unit-3

# Regression

# What is Regression ?

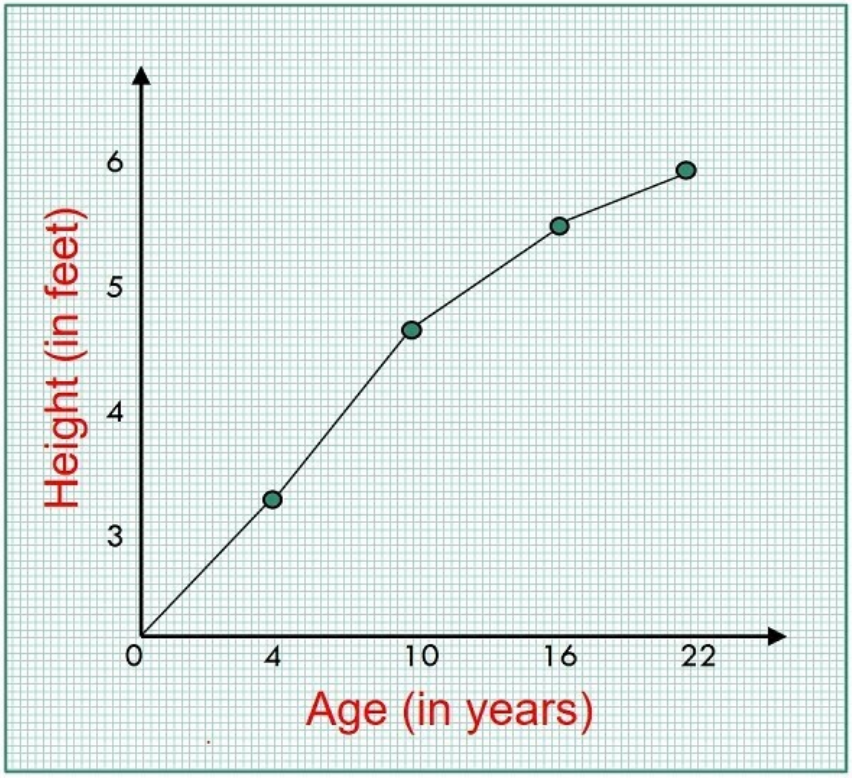
- Regression searches for relationships among variables.
- Regression analysis is a set of statistical methods used for the estimation of relationships between a dependent variable and one or more independent variables.
- It can be utilized to assess the strength of the relationship between variables and for modeling the future relationship between them.

https://realpython.com/linear-regression-in-python/

# independent & dependent variable

- The **independent variable** is the cause. Its value is **independent** of other variables in your study.
- The **dependent variable** is the effect. Its value **depends** on changes in the independent variable.

# Dependent Vs Independent Variables




Key Differences

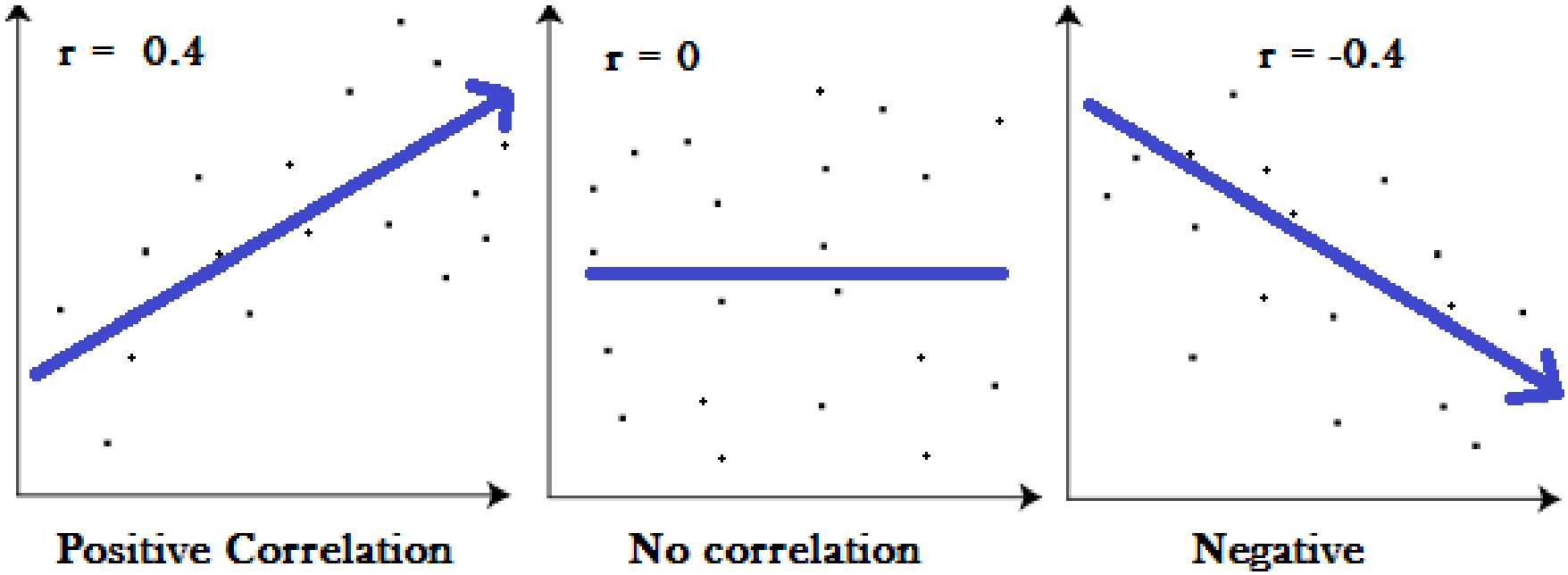
# Correlation Coefficient

- Correlation coefficients are used to measure the **strength of the relationship** between two variables.
- Pearson correlation is the one most commonly used in statistics. This measures the strength and direction of a linear relationship between two variables.

# Pearson Correlation Coefficient

- There are several types of correlation coefficients, but the one that is most common is the **Pearson correlation (*r*).** This measures the strength and direction of the linear relationship between two variables.

- Correlation coefficient formulas are used to find how strong a relationship is between data. The formulas return a value between -1 and 1
  - 1 indicates a strong positive relationship.
  - -1 indicates a strong negative relationship.
  - A result of zero indicates no relationship at all.


r = 0.4
Positive Correlation
r = 0
No correlation
r = -0.4
Negative

# Formula of Correlation

$$r = \frac{n(\Sigma xy) - (\Sigma x)(\Sigma y)}{\sqrt{[\, n\Sigma x^2 - (\Sigma x)^2 \,]\,[\, n\Sigma y^2 - (\Sigma y)^2 \,]}}$$

# Find Correlation Coefficient

| Serial No | Age (years)<br>X | Weight (Kg)<br>Y |
|---|---|---|
| 1 | 7 | 12 |
| 2 | 6 | 8 |
| 3 | 8 | 12 |
| 4 | 5 | 10 |
| 5 | 6 | 11 |
| 6 | 9 | 13 |

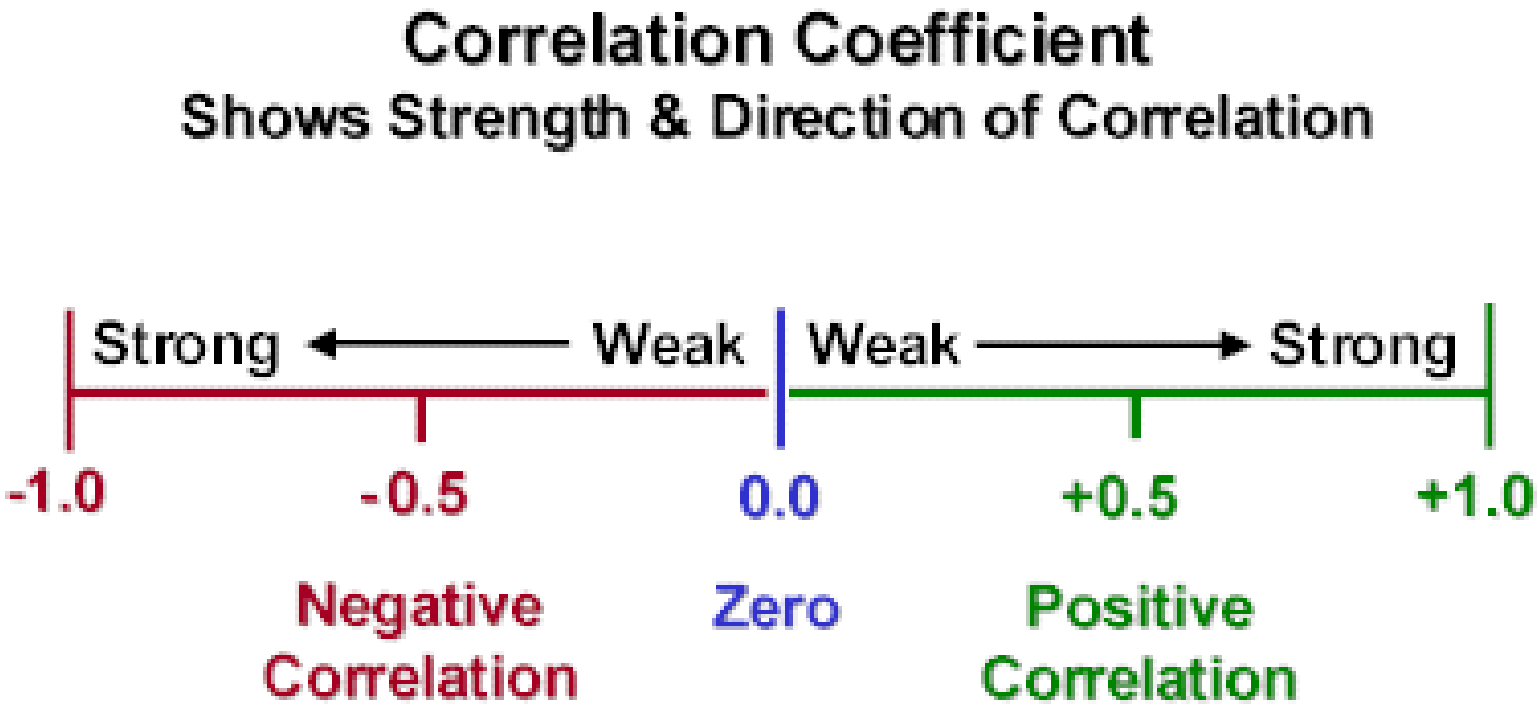
# Solution

| Serial n. | Age (years) (x) | Weight (Kg) (y) | xy | $X^2$ | $Y^2$ |
|---|---|---|---|---|---|
| 1 | 7 | 12 | 84 | 49 | 144 |
| 2 | 6 | 8 | 48 | 36 | 64 |
| 3 | 8 | 12 | 96 | 64 | 144 |
| 4 | 5 | 10 | 50 | 25 | 100 |
| 5 | 6 | 11 | 66 | 36 | 121 |
| 6 | 9 | 13 | 117 | 81 | 169 |
| Total | ∑x= 41 | ∑y= 66 | ∑xy= 461 | ∑x2= 291 | ∑y2= 742 |

**Answer r = 0.759**

# Result Discussion

- If $r$ = Zero  this means no association or correlation between the two variables.
- If $0 < r < 0.25$ = weak correlation.
- If $0.25 \leq r < 0.75$ = intermediate correlation.
- If $0.75 \leq r < 1$ = strong correlation.
- If $r = 1$ = perfect correlation.




http://www.edugyan.in/2017/02/correlation-coefficient.html

# Need of Regression

- Regression is a **statistical approach** used in finance, investment, and other fields to identify the **strength and type of a connection** between one dependent variable (typically represented by Y) and a sequence of other variables (known as independent variables).
- Regression is essentially the **"best guess"** at utilizing a collection of data to generate some form of forecast. It is the process of fitting a set of points to a graph.

# Applications of Regression

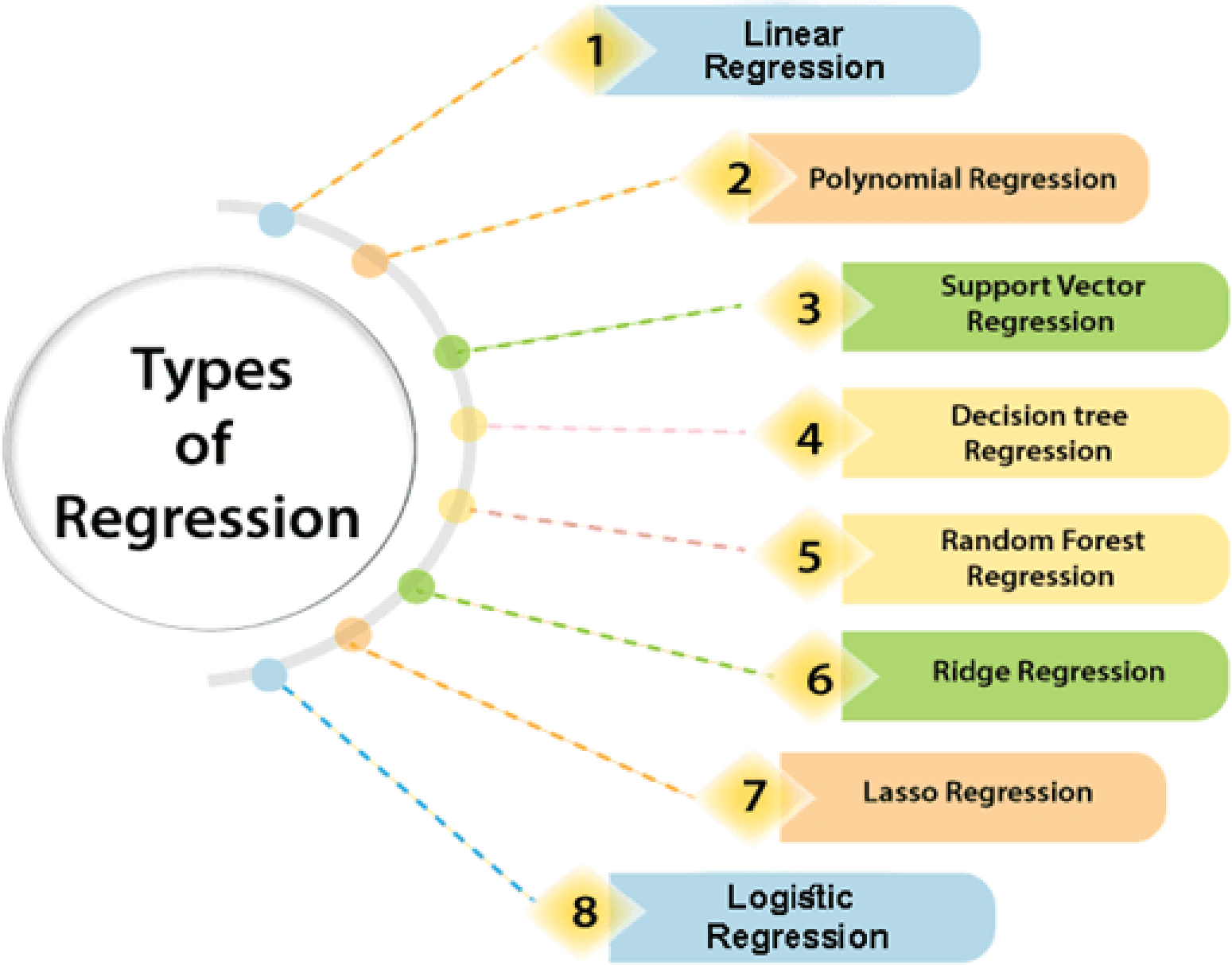
- Linear regression can be used in market research studies and customer survey results analysis
- Linear regressions can be used in business to evaluate trends and make estimates or forecasts.
- Linear Regression can be also used to assess risk in financial services or insurance domain. For example, a car insurance company might conduct a linear regression to come up with a suggested premium table using predicted claims to Insured Declared Value ratio.

# Applications of Regression

- Height and weight — as height increases, you'd expect the weight to increase, but not perfectly.
- Alcohol consumed and blood alcohol content — as alcohol consumption increases, you'd expect one's blood alcohol content to increase, but not perfectly.
- Vital lung capacity and pack-years of smoking — as the amount of smoking increases (as quantified by the number of pack-years of smoking), you'd expect lung function (as quantified by vital lung capacity) to decrease, but not perfectly.
- Driving speed and gas mileage — as driving speed increases, you'd expect gas mileage to decrease, but not perfectly.

# Types of Regression

- Simple Linear Regression
- Multiple Linear Regression
- Polynomial Regression


Types of Regression
1 Linear Regression
2 Polynomial Regression
3 Support Vector Regression
4 Decision tree Regression
5 Random Forest Regression
6 Ridge Regression
7 Lasso Regression
8 Logistic Regression

# Simple Linear Regression

- Simple linear regression is a statistical approach that allows us to study and summarize the relationship between two continuous quantitative variables.
- Out of the two variables, one variable is called the dependent variable, and the other variable is called the independent variable.
- Our goal is to predict the dependent variable's value based on the value of the independent variable.
- A simple linear regression aims to find the best relationship between X (independent variable) and Y (dependent variable).

Mathematically the relationship can be represented with the help of following equation

$$Y=A+Bx$$

**Y** is the dependent variable.
**X** is the independent variable.
**A** and **B** are constants which are called the coefficients.

$$A = \frac{(\Sigma y)(\Sigma x^2) - (\Sigma x)(\Sigma xy)}{n(\Sigma x^2) - (\Sigma x)^2}$$

$$B = \frac{n(\Sigma xy) - (\Sigma x)(\Sigma y)}{n(\Sigma x^2) - (\Sigma x)^2}$$

# Exercise - 1

| SUBJECT | AGE<br>X | GLUCOSE LEVEL<br>Y |
|---|---|---|
| 1 | 43 | 99 |
| 2 | 21 | 65 |
| 3 | 25 | 79 |
| 4 | 42 | 75 |
| 5 | 57 | 87 |
| 6 | 59 | 81 |

# Solution of Exercise - 1

| SUBJECT | AGE X | GLUCOSE LEVEL Y | XY | X2 | Y2 |
|---|---|---|---|---|---|
| 1 | 43 | 99 | 4257 | 1849 | 9801 |
| 2 | 21 | 65 | 1365 | 441 | 4225 |
| 3 | 25 | 79 | 1975 | 625 | 6241 |
| 4 | 42 | 75 | 3150 | 1764 | 5625 |
| 5 | 57 | 87 | 4959 | 3249 | 7569 |
| 6 | 59 | 81 | 4779 | 3481 | 6561 |
| Σ | 247 | 486 | 20485 | 11409 | 40022 |

From the above table, **Σx** = 247, **Σy** = 486, **Σxy** = 20485, **Σx²** = 11409, **Σy²** = 40022.
**n** is the sample size (6, in our case).

$y' = a + bx$

$\mathbf{y' = 65.14 + 0.385225x}$

# Exercise - 2

A sample of 6 persons was selected the value of their age ( x variable) and their weight is demonstrated in the following table. Find the regression equation and what is the predicted weight when age is 8.5 years.

| Serial no. | Age (x) | Weight (y) |
|---|---|---|
| 1 | 7 | 12 |
| 2 | 6 | 8 |
| 3 | 8 | 12 |
| 4 | 5 | 10 |
| 5 | 6 | 11 |
| 6 | 9 | 13 |

# Solution of Exercise - 2

| Serial no. | Age (x) | Weight (y) | xy | $X^2$ | $Y^2$ |
|---|---|---|---|---|---|
| 1 | 7 | 12 | 84 | 49 | 144 |
| 2 | 6 | 8 | 48 | 36 | 64 |
| 3 | 8 | 12 | 96 | 64 | 144 |
| 4 | 5 | 10 | 50 | 25 | 100 |
| 5 | 6 | 11 | 66 | 36 | 121 |
| 6 | 9 | 13 | 117 | 81 | 169 |
| Total | 41 | 66 | 461 | 291 | 742 |

$$Y = 4.675 + 0.92x$$